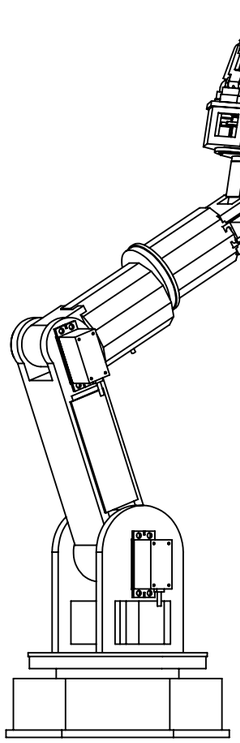
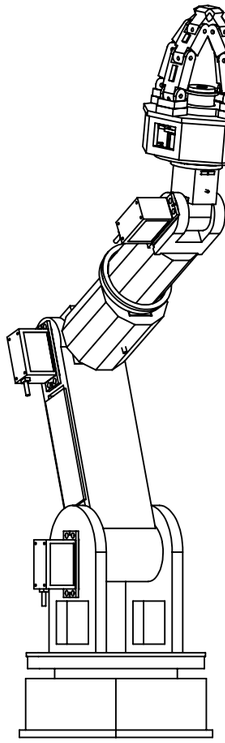
IRIS CLAW ROBOTIC ARM - GENERAL ARRANGEMENT

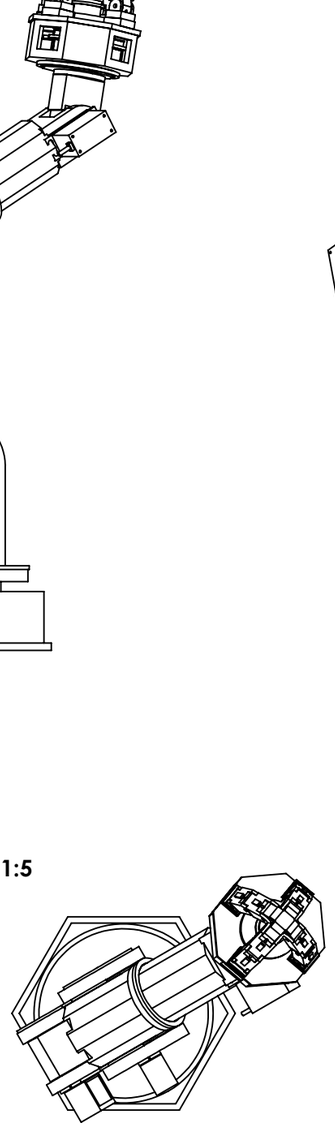
6-DOF arm + 4-finger swivel claw | 3D printed | 11 hobby servos | units: mm



FRONT 1:5



RIGHT 1:5



ISOMETRIC 1:4

TOP 1:5

ITEM NO.	DESCRIPTION	QTY.
1	Forearm lower - elbow hub, houses J4 roll servo	1
2	Middle segment - 26 mm, two tendon channels	4
3	Claw tip - hooked distal segment, tendon anchor	4
4	Proximal segment - 32 mm, two tendon channels	4
5	Palm hub - vented, 4 finger servos + swivel servo + camera	1
6	Finger mount - fork for the proximal knuckle pin	4
7	Swivel ring - moves fingers 2 and 4 up to 55 deg	1
8	Wrist link - J5 output, holds J6 hand-roll servo	1
9	J5 wrist pitch - MG996R 10 kg-cm	1
10	Forearm upper - rolls on J4, wrist fork + J5 seat	1
11	Accent ring - forearm roll joint	1
12	J4 forearm roll - MG996R 10 kg-cm	1
13	Upper arm - truss beam, shoulder hub to elbow fork	1
14	J3 elbow pitch - DS3218 20 kg-cm 270 deg	1
15	Upper-arm side panels - clip-on cosmetic covers	1
16	Accent ring - turntable (translucent, LED-ready)	1
17	Hex base - weighted foot, holds J1 yaw servo	1
18	Turntable yoke - J1 output, shoulder cheeks + J2 servo seat	1
19	J2 shoulder pitch - DS3235 35 kg-cm 270 deg	1
20	100 mm lazy-susan bearing - carries arm weight off J1	1
21	J1 base yaw - DS3218 20 kg-cm 270 deg	1
22	Accent ring - shoulder pin side	1
23	Accent ring - elbow pin side	1
24	J6 hand roll - MG90S 2.2 kg-cm	1
25	Accent ring - hand roll joint	1
26	Accent ring - camera bezel	1
27	Finger swivel - MG90S, drives swivel ring gear	1
28	Grip pad - soft TPU 95A, glued to claw	4
29	Finger drive - MG90S, pulls flex/extend tendon pair	4

NOTES
1. Print PLA, 3 walls, 25% gyroid; 100% infill near joints.
2. Pins: M3 bolts in fingers, M4 on idler side of J2/J3/J5.
3. Tendons: 0.5 mm braided line in PTFE, flex + extend pair.
4. Power: 6 V 10 A supply, PCA9685 driver, laptop via USB.
5. Orange = servo, teal = PETG ring, amber = claw / TPU pad.

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